THE UNIVERSITY OF AZAD JAMMU AND KASHMIR MUZAFFARABAD



Department of Software Engineering

Subject: Software Testing Lab

Lab Instructor: Engr. Fahad Nisar

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Lab Task:1

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1. Lab Objective

The objective of this lab is to install Node.js and npm, initialize a Node.js project, set up Playwright with JavaScript, and verify the setup by successfully running the generated sample test cases.

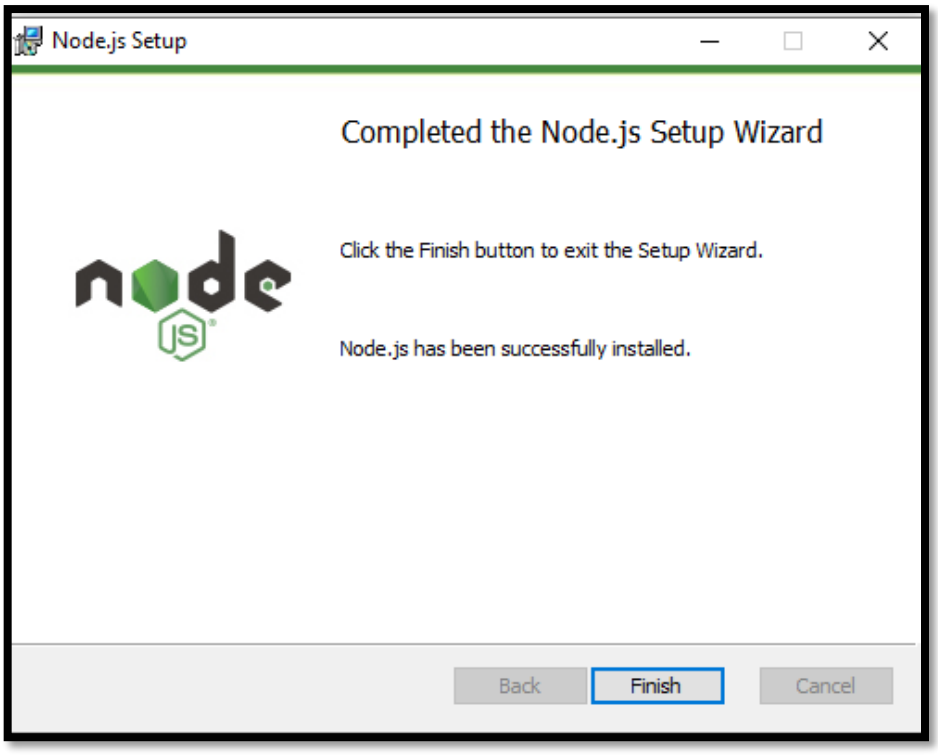
2. System Requirements & Environment

- **Operating System:** Windows 11
- **Internet Connection:** Active (required for package downloads and visiting online test targets)
- **Command Line Tool:** Command Prompt (CMD)

3. Step-by-Step Practical Implementation

Step 1: Install Node.js

1. Opened the official download page (`[https://nodejs.org/en/download](https://nodejs.org/en/download)`) in a browser.
2. Downloaded the Windows Installer (`.msi`) for the LTS (Long Term Support) release.
3. Executed the `.msi` setup wizard, accepted the end-user license agreement, kept all default options selected (including npm package manager and "Add to PATH"), and completed the installation.



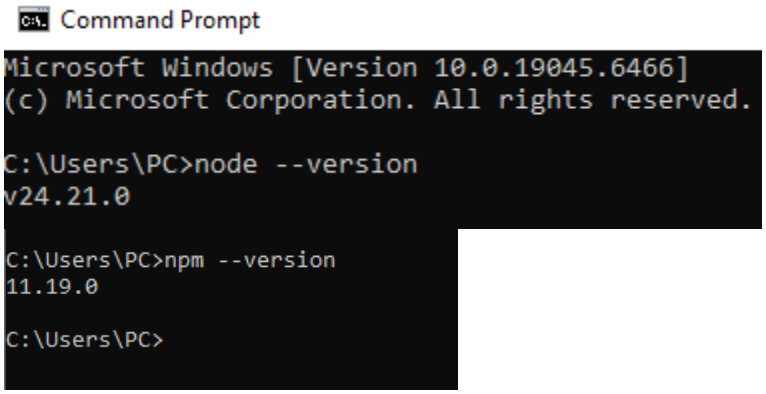
Step 2: Verify Node.js and npm Installation

1. Opened a new Command Prompt window to update the environment variables.
2. Executed the following commands to check the installed versions:

   DOS

   ```
   node --version
   npm --version
   ```

3. Confirmed that both commands returned their respective version numbers on the terminal screen.



Step 3 & Step 4: Project Folder Creation & Node.js Initialization

1. Created a dedicated working directory for the lab and navigated into it using the Command Prompt:

   Bash

   ```
   mkdir software-testing-lab
   cd software-testing-lab
   ```

2. Initialized a new Node.js project inside the folder with default configurations by executing:

   Bash

   ```
   npm init -y
   ```

```
C:\Users\PC>mkdir software-testing-lab

C:\Users\PC>cd software-testing-lab

C:\Users\PC\software-testing-lab>npm init -y
Wrote to C:\Users\PC\software-testing-lab\package.json:

{
  "name": "software-testing-lab",
  "version": "1.0.0",
  "description": "",
  "main": "index.js",
  "scripts": {
    "test": "echo \"Error: no test specified\" && exit 1"
  },
  "keywords": [],
  "author": "",
  "license": "ISC",
  "type": "commonjs"
}
```

Step 5 & Step 6: Playwright Setup and Project Inspection

1. Installed and initialized Playwright within the project folder by running:

   Bash

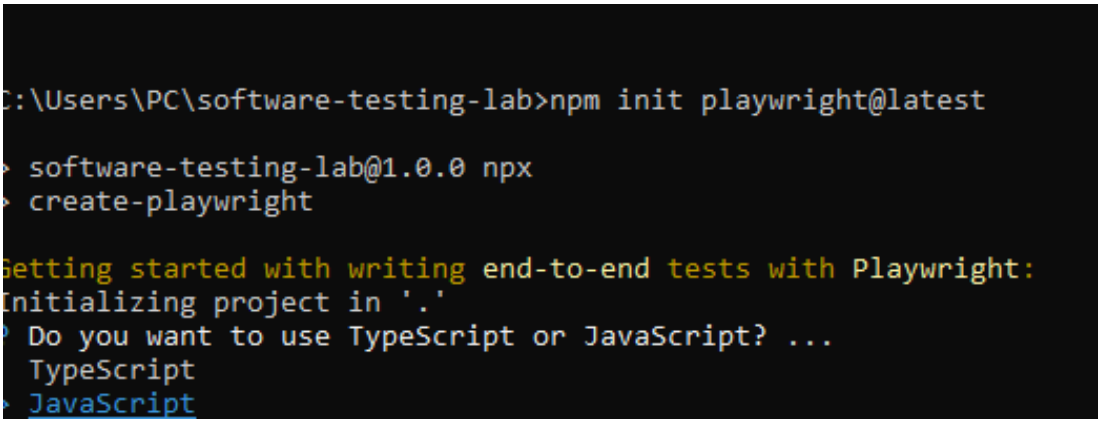
   ```
   npm init playwright@latest
   ```

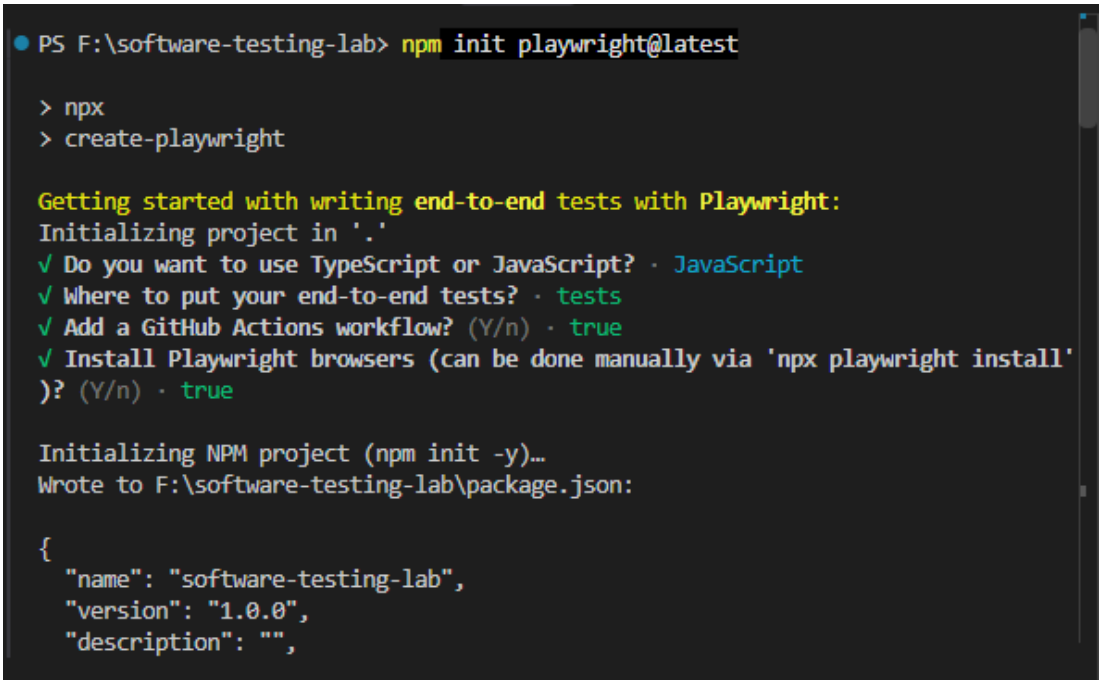
2. Configured the setup prompts as follows:
   - **Choose Language:** JavaScript
   - **Test Folder:** `tests`
   - **Add a GitHub Actions workflow:** No
   - **Install Playwright browsers:** Yes
3. Verified the installed Playwright version by executing:

   Bash

   ```
   npx playwright --version
   ```

4. Inspected the generated project structure, which includes `package.json`, `package-lock.json`, `node_modules`, `playwright.config.js`, and `tests/example.spec.js`.





Step 7: Run the Sample Tests

1. Executed the default generated sample test suite in headless mode:

   DOS

   ```
   npx playwright test
   ```

2. Ran the tests in headed mode using the Chromium browser to watch the browser execution visually:

   DOS

   ```
   npx playwright test --project=chromium --headed
   ```

```
PS F:\software-testing-lab> npx playwright test

  Running 6 tests using 2 workers
    6 passed (35.8s)

  To open last HTML report run:

    npx playwright show-report

PS F:\software-testing-lab> npx playwright show-report

    Serving HTML report at http://localhost:9323. Press Ctrl+C to quit.
```

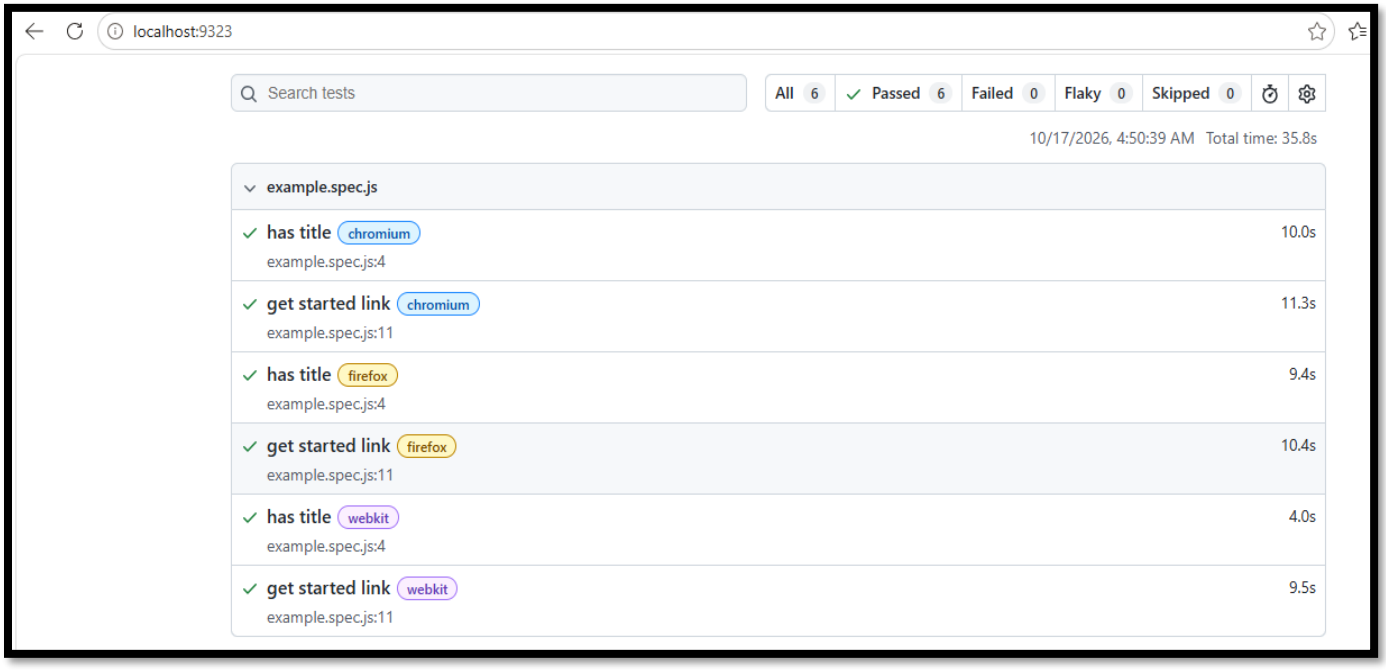
Step 8: View the HTML Report

1. Opened the automatically generated HTML report by running:

   DOS

   ```
   npx playwright show-report
   ```

2. Inspected the test details in the default web browser and pressed `Ctrl + C` in the Command Prompt to stop the local web server.



4. Conclusion

- Node.js and npm were successfully installed and verified on Windows 11.
- A Node.js project environment was configured and initialized with Playwright and JavaScript.
- All sample tests were executed successfully in both headless and headed modes, and results were reviewed via the Playwright HTML report.